Manish Desireddi

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SUMMARY

B.Tech Candidate in Artificial Intelligence & Data Science with a drive to engineer AI pipelines and computing architectures. Passionate about the intersection of Generative AI, Agentic AI, distributed systems, and hardware optimization (PC building). Adaptive to transforming theoretical machine learning/ deep learning concepts into scalable solutions, ranging from autonomous agents to complex e-commerce ecosystems. Attempting to leverage expertise in data analytics, interactive design, and full-stack engineering to solve complex technical challenges.

TECHNICAL EXPERIENCE

Nokia - Solutions and Networks Aug. 2025 – Present
Automation Engineer Bangalore, India

- Orchestrated test pipelines by engineering **Directed Acyclic Graphs (DAGs)** using **Apache Airflow** to automate test cases for Self-Optimising Networks (SON), significantly reducing execution time.
- Developed data ingestion pipelines to push telemetry and test results from DAGs directly into **Elasticsearch Database (ESDB)**, ensuring data availability.
- Integrated **Jenkins** automation to enforce system stability and continuous integration, creating a reliable testing environment that minimized regression risks.
- Designed dashboards in **Grafana**, transforming raw database metrics into actionable visual insights for the Performance and Verification team.
- Utilized **Robot Framework** for functional test automation and **LoadRunner** for performance analysis to ensure network resilience under load.

PROJECTS

Text to 3D | *Python, PyTorch, DL, GANs, LoRA, RAG, Flask, Celery, Redis, Docker* | *Generative AI*

- Architected a Deep Learning pipeline from scratch with **LoRA (Low-Rank Adaptation) adapters**, enabling **rapid self-training** on specific game asset styles without full-model retraining, bypassing pre-built APIs for full control over voxel density and texture mapping.
- Trained a GAN architecture over **100 epochs on the ShapeNet dataset**, utilizing a loss function and **gradient accumulation** to handle effective **batch sizes of 16**, achieving a **Fréchet Inception Distance (FID) score of 24.5**.
- Implemented a **Retrieval-Augmented Generation (RAG)** system using FAISS to retrieve topological constraints and texture maps from a local vector database, improving 3D mesh structural integrity by 35%.
- Achieved a **40% reduction in inference latency (<300ms per asset)** by implementing mixed-precision training (FP16) and optimizing the voxel discriminator.
- Built the backend using Flask and **Celery/Redis** to asynchronously manage heavy GPU compute loads during generation.

Windows Personal Assistant | *Python, Mistral-7B, Coqui XTTS, Distil-Whisper, Transformers, Win32 API, RabbitMQ, Electron* | *Agentic AI*

- Fine-tuned a **4-bit quantized Mistral-7B-Instruct model** to act as an OS agent, enabling natural language interpretation of complex system commands (File Mgmt, System Control) with 64% intent recognition accuracy.
- Integrated **Distil-Whisper** for low-latency Speech-to-Text (STT) and **Coqui XTTS v2** for high-quality Text-to-Speech, supporting local native languages (Hindi/Telugu) with minimal CPU overhead.
- Engineered a Local **RAG** system where the LLM contextualizes user queries against local file indices to provide accurate, privacy-centric replies without cloud dependency.
- Utilized **RabbitMQ** to decouple the voice processing pipeline from the execution engine, preventing UI freezes during heavy inference tasks on the local workstation.
- Implemented administrator-level execution safety rails, allowing the agent to perform autonomous tasks while maintaining system stability and security.

AI PC Building E-Commerce Platform | *React, FastAPI, Llama-3 (GGUF), Redis, Celery, LangChain, MongoDB*

- Developed a full-stack e-commerce platform for PC components, featuring a backend powered by a **Llama-3-8B (GGUF) LLM** to provide real-time, context-aware hardware recommendations.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** that dynamically fetches live pricing and compatibility data, allowing the LLM to generate up-to-date build guides instantly.
- Engineered a high-concurrency architecture using **FastAPI, Redis and Celery**, enabling the system to handle parallel inference requests locally without bottlenecking the main application thread.
- Created a frontend interface where user queries trigger live backend inference, delivering personalized component suggestions based on **performance benchmarks and budget constraints**.
- Optimized the entire LLM and database stack to run locally on a standard workstation, demonstrating efficient resource management for heavy AI workloads.

EDUCATION

KL University Expected May 2026
B.Tech in Artificial Intelligence & Data Science | ***CGPA: 8.2*** Guntur, AP

- **Specialization**: Game Development & UX Design

Tirumala Jr. Kalasala May 2022
Higher Secondary Education (Class XII) | ***92.8%*** Andhra Pradesh

SKILLS

AI & Machine Learning : Generative AI, Deep Learning, GANs, Agentic AI, Open Source LLMs, NLTK, spaCy
Languages : Python, C++, C#, C, R, SQL, JavaScript, HTML5, CSS3
Backend & Cloud : Flask, Django, Java (Spring Boot), REST APIs, Fast APIs, Azure, AWS, MongoDB Atlas
DevOps & Tools : Docker, Kubernetes, Celery, Redis, RabbitMQ, Git, Robot Framework, LoadRunner, Jira, Airflow, Jenkins, Grafana
Game Dev & Visualization : Unity, Unreal Engine, Three.js, WebGL
Frontend : React.js, Tailwind CSS, Electron, Tauri
OS : Windows 10/11, Linux, Ubuntu 22.04LTS, Fedora

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate

Oracle Cloud Infrastructure 2024 Generative AI Professional (1Z0-1127-24)

Oracle Cloud Infrastructure 2025 DevOps Professional (1Z0-1109-25)

Salesforce Certified AI Associate

Unity Certified User Programmer

IBM Python Certification

AI Algorithms for Gaming(LinkedIn)

LEADERSHIP & ACHIEVEMENTS

Vice President - Student Body (KL-SAC) Academic Year 2024-25
KL University Guntur, India

- **Strategic Leadership Promotion:** Fast-tracked to Vice President from HR & Communications due to exemplary performance in organizational restructuring and student engagement strategies.
- **Accreditation Representation (NAAC):** Spearheaded student representation during the National Assessment and Accreditation Council (NAAC) inspection; presented key engagement metrics that directly contributed to the institution's quality assurance benchmarks.
- **Operational Management:** Orchestrated administrative operations for university-wide technical symposiums, serving as the primary liaison between university management and diverse student factions to streamline policy execution.

- **Crisis and Event Management:** Directed cross-functional teams to manage logistics for large-scale campus events, ensuring compliance with institutional standards while fostering an inclusive campus culture.

Additional Initiatives

- **Mathematics Educator (Social Internship):** Conducted training modules for the International Mathematics Olympiad (IMO) at Kendriya Vidyalaya, simplifying complex number theory for underprivileged students.
- **Production Operations (Mrs. India 2025):** Optimized live production schedules and minimized technical equipment downtime through strategic resource allocation and real-time troubleshooting.